In Llama, Saying "I Don't Know" Tracks Entity Popularity More Than Knowledge

Anonymous Author(s)
Affiliation
Address
email

Abstract

When a language model answers a factual question with "I don't know", is it reporting that it lacks the answer? We cross entity popularity (Wikipedia page views in PopQA) with knowledge labels taken from the model's own sampled answers. On obscure entities, Llama-3.1-8B-Instruct says "I don't know" to 43% of questions it answers correctly in at least 9 of 10 forced samples; on popular entities the rate is 0%, although its answer entropy is similar in the two cases. The gap holds with knowledge relabelled from a second prompt, with unknown facts regraded by a larger model, and when only replies that name no answer count. Gemma-2-9B rarely abstains under a plain prompt, but when the prompt permits "I don't know" its abstention also tracks popularity more than knowledge. As a control, answer entropy follows knowledge, as the labelling implies. Steering and logit-lens readouts do not isolate a familiarity mechanism.

1 Introduction

A model that says "I don't know" when it lacks a fact avoids the worst hallucinations [Kalai et al., 2025]. Confidence estimates usually read uncertainty over answers [Kadavath et al., 2022, Kuhn et al., 2023, Stolfo et al., 2024], models use such confidence when a prompt offers abstention [Kumaran et al., 2026], and verbal and answer-level uncertainty diverge [Ji et al., 2025]. What each follows is less clear. For work on representational convergence [Huh et al., 2024], the question is whether models that agree on answer uncertainty also agree on what makes them abstain; we can ask it only where abstention is measurable, here Llama and, under a permissive prompt, Gemma.

People's sense that they can answer a question depends on how familiar its terms are as well as on whether the answer can be retrieved [Reder and Ritter, 1992, Schwartz and Metcalfe, 1992, Koriat and Levy-Sadot, 2001]. In language models, "known entity" features suppress a default "can't answer" response, and Lindsey et al. [2025] propose that a familiar name can trigger them when the fact is missing, so that the model answers when it should abstain; we call this a *misfire*. The converse failure is an unfamiliar name that does not suppress the default, so that the model abstains on a fact it knows. Tests of the idea define an entity as known when the model answers about it correctly [Ferrando et al., 2025], so familiarity and knowledge move together. Familiarity has been separated from factual reliability in representations [Brzezinka, 2026a] and shown to shape hallucination through fine-tuning data [Kang et al., 2025], but not crossed with knowledge in abstention behaviour.

We cross them. From PopQA [Mallen et al., 2023] we take questions about obscure and popular entities and label each fact known or unknown from the model's own answers

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[Gekhman et al., 2024]. If abstention follows familiarity, two cells show it: popular unknown facts should draw few "I don't know" replies (misfires) and obscure known facts many (the default not suppressed). Our contribution is this popularity-versus-knowledge dissociation in abstention, its checks, and a comparison across families of what abstention and answer uncertainty follow (Sections 3.1 and 3.2); steering and lens analyses then test, and fail to isolate, a mechanism (Section 3.3).

2 Setup

Questions and labels. We use the 16 relations of PopQA. Within each relation, subjects in the bottom and top deciles of Wikipedia page views, a proxy for familiarity, form the *obscure* and *popular* arms. For each model, a greedy answer and ten samples ($T = 0.7$) under a *forced-answer prompt* (the chat template plus the assistant prefix "The answer is") are graded against gold aliases: a fact is *known* if the greedy answer and at least nine samples are correct and *unknown* if no sample is; the band between is dropped. This leaves 2,303, 2,392, 2,282 and 2,378 questions for Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, Gemma-2-9B-it and Mistral-7B-Instruct-v0.3 [Grattafiori et al., 2024, Qwen et al., 2024, Team et al., 2024, Jiang et al., 2023] (cell counts in Appendix A).

Two readouts. *Answer uncertainty* is the first-token entropy under the forced-answer prompt. Because the knowledge labels are sampled from the same distribution, entropy is expected to follow knowledge; it serves as a control, and we also relabel knowledge from an independent prompt. *Abstention* needs the option to decline, so it is measured with a *free prompt*, the chat template alone: for up to 100 questions per popularity × knowledge cell (387 in total) we draw three replies ($T = 0.7$, up to 96 tokens). Two LLM judges (Qwen2.5-32B-Instruct, Gemma-2-27B-it) label each reply DONT_KNOW, CLARIFY (asks which entity is meant) or ANSWER, and a reply abstains when both say DONT_KNOW. On Llama they agree at $\kappa = 0.62$ [0.57, 0.67], above a pre-specified bar of 0.60; on the other families they do not (Section 3.2). No human validated the judges.

Effects. The popularity effect is the obscure-minus-popular difference in a readout averaged over the two knowledge levels; the knowledge effect is the unknown-minus-known difference averaged over popularity. Their difference, popularity effect minus knowledge effect, is the model's *popularity preference*: positive when abstention follows popularity more than knowledge. In this balanced 2×2 it equals abstention on obscure known facts minus abstention on popular unknown facts. Intervals are 95% percentile bootstraps over questions within cells, not clustered by relation. Unless marked post hoc, every analysis followed a plan written and hashed before it ran (*pre-specified*; Appendix G).

3 Results

3.1 In Llama, abstention tracks popularity; answer uncertainty follows knowledge

Llama says "I don't know" to 0.43 of replies about obscure entities whose facts it knows and to none about popular entities whose facts it knows, at forced-answer entropies of 0.95 and 0.86 nats (Table 1, Figure 1). Over the design the popularity effect on abstention is +0.39 [+0.33, +0.45] and the knowledge effect +0.09 [+0.03, +0.15], a popularity preference of +0.30 [+0.20, +0.40]. It keeps its sign under five other measures of abstention, from +0.13 to +0.44 (Table 2); the primary measure was designed after the smallest of these was seen. The control behaves as expected: entropy rises +2.14 nats when Llama lacks the fact and +0.51 when the entity is obscure.

Are the knowledge labels right? Relabelling knowledge with a paraphrased question and prefix agrees with the original labels at $\kappa = 0.79$. Of the 87 obscure known facts, 27 stay known and abstain at 0.46; the preference is +0.32 [+0.16, +0.49], and entropy still follows knowledge. Popular entities may have valid answers outside PopQA's aliases, so Llama-3.3-70B regraded forced answers by meaning: it found 9 of the 100 popular unknown facts known and 31 in between, and with its labels the preference is +0.23 [+0.12, +0.35].

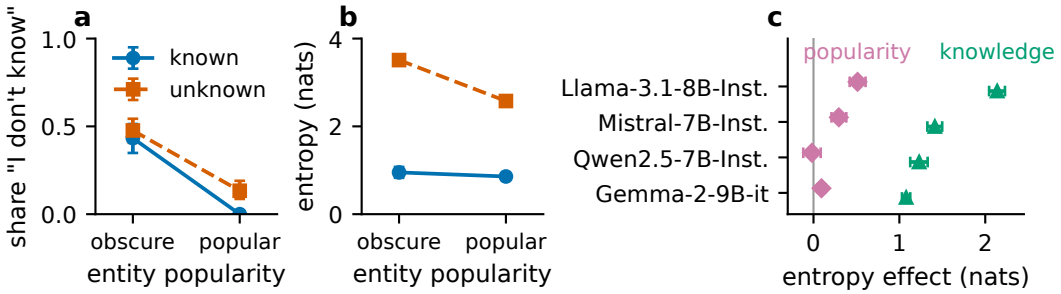


Figure 1: **In Llama, saying "I don't know" tracks popularity, while answer uncertainty follows knowledge.** (a) Share of Llama's free replies both judges label "I don't know"; (b) its first-token entropy when the reply must begin "The answer is" (values and n in Table 1); (c) popularity and knowledge effects on that entropy in four instruct models. 95% bootstrap intervals.

Table 1: Llama-3.1-8B-Instruct by cell: share of free replies both judges label "I don't know", and forced-answer first-token entropy, with 95% intervals and question counts.

	Obscure		Popular	
	known	unknown	known	unknown
Says "I don't know"	0.43 [0.35, 0.51]	0.48 [0.41, 0.55]	0.00 [0.00, 0.00]	0.13 [0.08, 0.19]
questions	87	100	100	100
Entropy (nats)	0.95 [0.83, 1.08]	3.51 [3.46, 3.56]	0.86 [0.80, 0.92]	2.58 [2.47, 2.69]
questions	87	1,179	676	361

Are the "I don't know" replies refusals? With Llama-3.3-70B as a third judge and a majority rule, $\kappa = 0.74$ and the preference is +0.44. At most 0.03 of the "I don't know" replies in any cell contain the gold answer (post hoc), and counting only replies the 70B judge labels as declining without naming any candidate gives +0.43 [+0.31, +0.54].

Does Llama fail to recognise obscure entities? It describes the entity correctly in 0.83 of its completions for obscure known facts and 0.75 for popular known ones, and adding this LLM-graded recognition score to a regression shrinks the popularity coefficient by 5%.

Is it ambiguity? If obscure names were ambiguous, Llama might hedge for that reason. The data point the other way (post hoc): popular names are shared with more Wikidata entities than obscure ones (median 11–13 against 1 for obscure known facts), and obscure known facts whose name is unique still draw "I don't know" in 0.43 of replies. How the name is spelled explains part of the gap between matched pairs of known facts, but barely changes the popularity coefficient over all questions (post hoc; Appendix A).

Is it familiarity? Neither model-side measure shows the obscure names to be unfamiliar to Llama: name surprisal is uncorrelated with page views (Appendix F) and recognition is higher for obscure known facts. The result has the shape the misfire idea predicts but does not establish its mechanism.

Does the split need two prompts? At the forced-answer position, the first token's lean towards hedge tokens (tokens that begin a refusal, such as " I", " not", " unknown") follows popularity and knowledge equally (−0.12 [−0.72, +0.45]; Appendix B). After seeing this we specified a reading at the free prompt: there the log-odds of a first token "I" follow popularity (+8.4 against +3.1) while the first token's entropy follows knowledge (+0.16 against +0.05 nats).

3.2 Other families: Gemma tracks popularity too when it may say "I don't know"

As expected, entropy follows knowledge in all four families (Figure 1c). Under the free prompt, Qwen, Gemma and Mistral rarely say "I don't know" and the judges disagree on them; with a third judge Gemma reaches $\kappa = 0.64$ but abstains in at most 0.11 of replies

in any cell, so its preference cannot be estimated. A pre-specified four-family test on the forced-answer hedge readout (hedging follows popularity more, entropy follows knowledge) found both in Gemma and one in Qwen and Mistral, but that readout shows no preference in Llama either (Appendix B). When the question ends "If you do not know the answer, say that you do not know", both models abstain often and three judges agree ($\kappa = 0.71$, 0.67). The popularity preference is then +0.47 [+0.35, +0.59] for Llama and +0.30 [+0.19, +0.40] for Gemma, which says "I don't know" to 0.41 of obscure known questions and 0.00 of popular known ones. Agreement between models on the forced-answer readout beyond shared labels met a post hoc rule for one of three pairs (Appendix E).

3.3 Mechanistic tests do not isolate a familiarity cause

Steering. As work steering refusal along recognition directions does [Ferrando et al., 2025, Brzezinka, 2026b], we fitted obscure-minus-popular, unknown-minus-known and abstain-minus-answer directions on half of the questions [Arditi et al., 2024, Rimsky et al., 2024], added each at a multiple (the dose) of the obscure-minus-popular direction's norm, chose the dose on a second set and scored a third (Appendix C). Obscure-minus-popular raised Llama's abstention from 0.01 to 0.10, +0.09 [+0.06, +0.12] over random directions and short of the pre-specified bar of +0.10, and Gemma's from 0.00 to 0.04 at 4× the norm (Appendix C, Figure 2a). In Llama it is nearly parallel to the abstention direction (cosine 0.95), partly by construction, since abstaining questions are mostly obscure; removing that component leaves +0.01 [−0.00, +0.02] (Gemma +0.02 [+0.01, +0.03]). Fitted on obscure questions alone, the abstention direction still has cosine 0.91, and removing it leaves +0.01 [+0.00, +0.03] (three judges). Obscure-minus-popular beats unknown-minus-known on abstention (+0.06 [+0.02, +0.09]), as the cells' rates predict, while in Gemma unknown-minus-known raises entropy +0.66 [+0.58, +0.74] nats more (Figure 2b). Steering thus moves a generic abstention signal [cf. Che and Li, 2026].

Lens. Does a popularity-sensitive hedge signal exist before instruction tuning? With prefitted Jacobian lenses [Gurnee et al., 2026], read at the forced-answer position in a pre-specified mid-depth window, popularity raises the hedge-token term in Llama-3.1-8B, Gemma-2-9B, Gemma-2-2B and OLMo-3-7B [Olmo et al., 2025], and knowledge raises it about as much (Appendix D, Figure 3). Instruct partners lean further towards knowledge, so this readout does not show where Llama-Instruct's preference arises.

4 Conclusion and limitations

In this extreme-contrast design, Llama's answer uncertainty follows knowledge while its abstention tracks popularity more, so in Llama answer uncertainty alone predicts abstention poorly. Of the two models whose abstention we could measure, both track popularity more than knowledge when permitted to abstain; whether other families share this is open.

Abstention is judged by LLMs not validated against human labels. The design compares only the extreme page-view deciles, keeps only facts answered correctly at least nine times in ten or never, and caps cells at 100 questions; models have 2–9B parameters and questions name a single entity. Hedging driven by ambiguity that the judges do not label as clarification is not excluded.

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A Robustness of the Llama abstention result

Design counts. Llama's pool has 87 known, 85 intermediate and 1179 unknown obscure facts, and 676, 285 and 361 popular ones; the two-judge sample is a stratified random sample of 100 questions per cell (seed 0), or all of a smaller cell. Intervals for differences of cell means use 10,000 within-cell bootstraps (seed 0); linear probability models use 2,000 bootstraps over questions.

Measurements. The single corrected judge gives the smallest difference. Its 24-token replies are often cut off before the model commits; why it and the regex disagree on the same replies is not established. The two-judge measurement with 96 tokens was designed for that reason after the +0.13 was seen and made primary before it ran. Regex recall on the 96-token replies, against the two-judge labels, is 0.99. Each judge's agreement with the three-judge majority: Qwen 0.98, Gemma 0.84, Llama-3.3-70B 1.00.

Independent labels (pre-specified). Knowledge was relabelled with one paraphrased question per relation, the assistant prefix "Answer:" and fresh samples, under the same rule. Of the 253 questions labelled under both prompts, 62 are known under both, 169 unknown under both, 15 known only originally and 7 only under the paraphrase (agreement 0.91, $\kappa = 0.79$); most other changes move popular known facts to the dropped band. Where both labellings agree, cells have 99 / 27 / 70 / 35 questions (obscure unknown, obscure known, popular unknown, popular known). Under the new labels, the popularity effect on abstention is +0.38 [+0.29, +0.47] and the knowledge effect +0.06 [−0.03, +0.15], and original-prompt entropy follows them (knowledge minus popularity +1.40 [+1.02, +1.73]).

Table 2: Llama's popularity preference under other measures of abstention and other knowledge labels, and the coefficient of the obscure arm under regression controls. Positive values mean abstention follows popularity more than knowledge. "Corrected": regex rates on 24-token replies corrected for the recall and precision of one LLM judge (Qwen2.5-32B-Instruct), measured on a subset labelled by that judge, not by people. Plan: pre-specified (pre) or post hoc. n: questions.

Variant	Replies	n	Plan	Estimate
Measures of abstention: popularity preference				
Two judges, both DONT_KNOW (primary)	3, 96 tokens	387	pre	+0.30 [+0.20, +0.40]
Two judges, DONT_KNOW or CLARIFY	same	387	pre	+0.31 [+0.21, +0.41]
Three judges, majority	same	387	pre	+0.44 [+0.32, +0.55]
70B judge, declines naming no candidate	same	387	pre	+0.43 [+0.31, +0.54]
Regex detector	same	387	post hoc	+0.43 [+0.31, +0.55]
Regex detector	20, 24 tokens	2,303	pre	+0.44 [+0.34, +0.53]
One LLM judge, corrected	same 24-token	2,303	pre	+0.13 [+0.03, +0.24]
Other knowledge labels (two judges): popularity preference				
Relabelled with a paraphrased prompt	3, 96 tokens	253	pre	+0.32 [+0.16, +0.49]
where both labellings agree	same		post hoc	+0.31 [+0.15, +0.48]
Unknown facts regraded by meaning	same		pre	+0.23 [+0.12, +0.35]
Popular unknown with the answer in a reply dropped	same		post hoc	+0.19 [+0.07, +0.31]
Guessable answers removed	same		post hoc	+0.30 [+0.19, +0.41]
Regression controls (two judges): coefficient of the obscure arm				
Relation fixed effects and knowledge only	3, 96 tokens	387	pre	+0.35
plus entity recognition	same	387	pre	+0.34 [+0.27, +0.41]
plus namesake count	same		post hoc	+0.34 [+0.27, +0.41]

Regraded unknown facts (pre-specified). Llama-3.3-70B graded each of a fresh greedy and ten sampled forced answers for every unknown fact as naming the gold entity or not. Of the 100 popular unknown facts in the sample, 60 stay unknown, 9 become known and 31 intermediate (dropped); of the 100 obscure unknown facts, 86, 2 and 12. Dropping instead every popular unknown question with a gold alias in any free reply (0.48 of them) gives +0.19 [+0.07, +0.31].

Strict declines (pre-specified). The 70B judge labelled each reply as declining without naming a candidate, hedging with a candidate, answering, or asking which entity is meant. Declines occur in 0.66 of obscure known replies and 0.00 of popular known ones; among replies both judges call DONT_KNOW, at most 0.03 per cell are hedged answers.

Entity recognition (pre-specified). Llama completed "{subject} is" after the question "What is {subject}?" (greedy and five samples), and Llama-3.3-70B graded each completion against the subject's English Wikidata description; recognition is the share of completions graded correct. Its correlation with log page views over all graded questions is $\rho = +0.52$. In the linear probability model its coefficient is -0.03 [-0.06, $+0.00$] per standard deviation, and it shrinks the popularity coefficient by 5% on the two-judge measure and 3% on the regex measure.

Guessable answers (post hoc). An answer counts as guessable if a gold alias of at least three characters appears in the subject's name, or if it is its relation's most common answer and covers at least 20% of the relation's pool. Among unknown facts alone the popularity gap in abstention is +0.34 [+0.26, +0.43]. On the regex measure over the 2,106 non-guessable questions the difference is +0.48 [+0.36, +0.59].

Namesakes (post hoc). For each subject we counted Wikidata entities with the same English label; medians are 1 (obscure known), 5 (obscure unknown), 11 (popular known) and 13

(popular unknown). The covariate is the standardised log count. The 73 obscure known facts with a unique name abstain at 0.43.

Free replies (post hoc). In the obscure known cell, 0.53 of free replies contain a gold alias, and 0.72 of the replies that do not abstain; for 24 of 87 questions, one free reply gives the correct answer and another says "I don't know". A linear probability model of abstention gives popularity +0.29 [+0.23, +0.36], knowledge −0.05 [−0.14, +0.03] and +0.14 per standard deviation of forced-answer entropy.

Name spelling (post hoc). Between 55 matched pairs of known facts, one obscure and one popular, the regex abstention gap of +0.76 shrinks to +0.43 [+0.18, +0.69] once the names' word frequency, token count and token rarity are held equal. Over the 2,303 labelled questions, the popularity coefficient on the regex measure moves from −1.87 to −1.79 logits per standard deviation after adding the name's word frequency, token count and token rarity. The matched pairs compare only known facts, one popular with one obscure fact of similar surface form, which is why they lose more of the gap.

B Abstention and the forced-answer hedge readout in all four families

Table 3: Judge agreement and abstention. Top: judge agreement and regex recall. Middle: abstention in the obscure, unknown cell. Bottom: three-judge popularity preference, then effects on first-token hedge-versus-answer log-odds at the forced-answer position, and the pre-specified four-family test, whose two conditions are that hedging follows popularity more than knowledge on this readout and that entropy follows knowledge more than popularity.

	Llama	Qwen	Gemma	Mistral
Two judges, κ (bar 0.60)	0.62	0.30	0.47	0.39
Three judges, Fleiss κ (bar 0.60)	0.74	0.41	0.64	0.54
Regex recall against two judges	0.99	0.58	0.48	0.73
Both judges DONT_KNOW	0.48	0.05	0.05	0.10
Both judges DONT_KNOW or CLARIFY	0.65	0.32	0.22	0.25
Three judges, majority	0.75	0.10	0.11	0.19
Regex detector	0.85	0.18	0.07	0.21
Three judges: popularity − knowledge	+0.44	–	+0.01	–
Hedge log-odds: popularity	+2.93	+1.14	+1.79	+2.48
Hedge log-odds: knowledge	+3.05	+3.63	−0.32	+2.08
Hedge log-odds: popularity − knowledge	−0.12	−2.49	+2.11	+0.40
Four-family test (pre-specified): conditions met	reference	one holds	both hold	one holds

Intervals for the popularity-minus-knowledge rows: three judges, Llama [+0.32, +0.55], Gemma [−0.03, +0.06]; hedge log-odds, Llama [−0.72, +0.45], Qwen [−3.41, −1.64], Gemma [+1.79, +2.42], Mistral [−0.55, +1.34]. The two judges that agree best are Qwen2.5-32B and Llama-3.3-70B ($\kappa = 0.96$ on Llama).

Permissive prompt (pre-specified). The user message was the question followed by "If you do not know the answer, say that you do not know.", on each model's existing two-judge question set, with three replies scored by the three judges and a majority rule. Cell rates (obscure unknown, obscure known, popular unknown, popular known): Llama 0.84, 0.69, 0.22, 0.00; Gemma 0.54, 0.41, 0.11, 0.00. The criterion let Gemma's verdict replace "cannot be estimated" only if its obscure unknown rate reached 0.25.

C Steering details

Pre-specified (Appendix G). Directions were fitted on a random half of each model's questions at layer round$(0.55 \times (L-1))$, the last position of the free prompt: obscure minus popular and unknown minus known (differences of cell means, each averaged over the other factor), and abstain minus answer (questions whose stored free-generation regex abstention rate is at

least 0.5 minus those with rate 0). Their natural norms are 4.8, 3.1 and 7.0 in Llama and 62.4, 66.5 and 105.3 in Gemma. The obscure-minus-popular direction with its abstention component projected out was rescaled to the original norm. Each direction was added at a dose $k \in \{0.5, 1, 2, 4, 8\}$ times the obscure-minus-popular direction's natural norm, at the last prompt position and every generated position. k was chosen on 150 popular known questions as the smallest value at which that direction beat five random directions by 0.10 in regex abstention, a rule separate from the confirmatory bar (1× for Llama, 4× for Gemma, whose obscure-minus-popular regex rate was 0.00 at 1× and 0.02 at 2×). All conditions were scored on 150 further popular known questions, 21 of which are also in the two-judge sample.

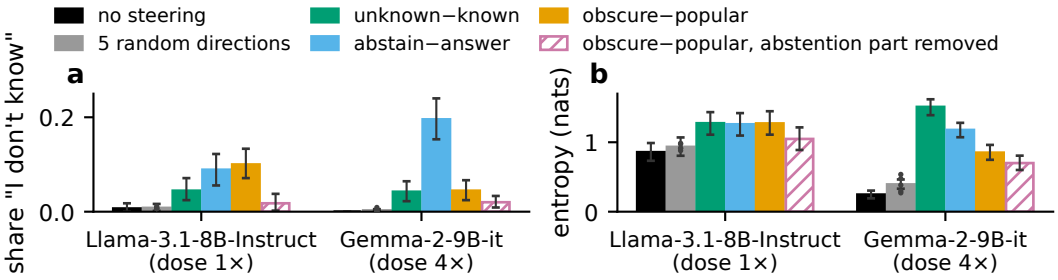


Figure 2: **In Llama, steering along the obscure-minus-popular direction adds little abstention beyond its overlap with an abstention direction.** (a) Share of free replies both judges label "I don't know" on 150 held-out popular questions each model knows; dots: random directions; hatched: obscure-minus-popular with its abstain-minus-answer component projected out. Gemma's two judges agree below the bar; three-judge rates in Appendix C. (b) First-token entropy under the forced-answer prompt. Error bars: 95% bootstrap over questions.

Table 4: Share of steered replies both judges label "I don't know" (regex detector in parentheses), 150 evaluation questions, 3 replies each.

	None	Random	Unk.−known	Abstain−answer	Obscure−popular	... abstention removed
Llama	0.01 (0.03)	0.008	0.04 (0.10)	0.09 (0.25)	0.10 (0.23)	0.02 (0.02)
Gemma	0.00 (0.00)	0.003	0.04 (0.03)	0.20 (0.74)	0.04 (0.27)	0.02 (0.07)

Primary scorer (both judges), obscure-minus-popular minus random: Llama +0.09 [+0.06, +0.12], Gemma +0.04 [+0.02, +0.06]; with abstention removed: +0.01 [−0.00, +0.02] and +0.02 [+0.01, +0.03]; obscure-minus-popular minus unknown-minus-known: +0.06 [+0.02, +0.09] and +0.00 [−0.03, +0.03]. Secondary scorer (a majority of three judges, added by amendment before judging): obscure-minus-popular minus random +0.17 [+0.13, +0.22] and +0.11 [+0.08, +0.15]; with abstention removed +0.02 [+0.00, +0.04] and +0.05 [+0.03, +0.07]. For Gemma the secondary scorer is the defensible one, since its two judges agree below the bar. The cosine between the obscure-minus-popular and abstention directions is 0.95 in Llama and 0.63 in Gemma. No condition produced degenerate text.

Within-level abstention direction (pre-specified). To remove the built-in overlap, the abstain-minus-answer direction was refitted on obscure questions only (587 abstaining against 55 answering in Llama, 36 against 520 in Gemma), with the same splits, layer and dose, and scored by a majority of three judges. Its cosine with obscure-minus-popular is 0.91 in Llama and 0.47 in Gemma. Rates for no steering, the random mean, obscure-minus-popular, the within-level abstention direction and obscure-minus-popular with that direction removed: Llama 0.02, 0.02, 0.19, 0.22, 0.03; Gemma 0.00, 0.01, 0.11, 0.41, 0.03. Over random directions, obscure-minus-popular adds +0.17 [+0.12, +0.22] and +0.10 [+0.07, +0.14], and with the within-level component removed +0.01 [+0.00, +0.03] and +0.03 [+0.01, +0.04]; by Llama's pre-specified rule the effect is carried by abstention.

D Lens: window effects, the instruct models and onsets

Pre-specified. Hedge term: log-sum-exp of the lens log-probabilities of the hedge tokens " not", " I", " unknown", " unclear" and " no"; the full readout subtracts the same quantity for the answer-opening tokens " a", " the", " that", " in", " Yes" and " yes". Window: relative depth $\ell/(L-1) \in [1/3, 2/3]$. Intervals: 2,000 within-cell bootstraps. Each model was labelled known or unknown on its own prompt with the same sampling rule.

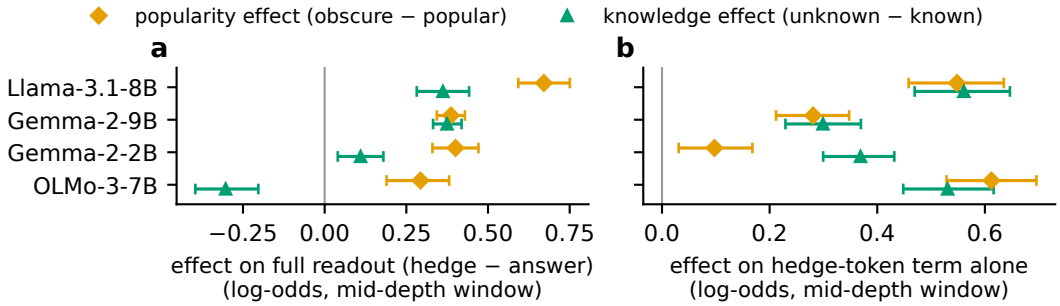


Figure 3: **In base models, the decoded hedge-token term responds to popularity and about as much to knowledge.** Jacobian-lens readout at the last position of "Question: ... Answer: The answer is", averaged over the pre-specified mid-depth window: (a) log-odds of five hedge tokens (e.g. " not", " unknown") against six answer-opening tokens (e.g. " the", " a"); (b) the hedge-token term alone. On (a), popularity exceeds knowledge in three models because the answer-token term also moves; on (b), knowledge is about as large or larger, except in OLMo, where it is slightly smaller with overlapping intervals. Points: popularity and knowledge effects with 95% bootstrap intervals. Lenses are fitted per model, so magnitudes are not comparable across models.

Table 5: Mid-depth window effects in base models, with 95% intervals.

	Full readout		Hedge term alone	
Model	Popularity	Knowledge	Popularity	Knowledge
Llama-3.1-8B	+0.67 [+0.59, +0.75]	+0.36 [+0.28, +0.44]	+0.55 [+0.46, +0.64]	+0.56 [+0.47, +0.65]
Gemma-2-9B	+0.39 [+0.34, +0.43]	+0.37 [+0.33, +0.42]	+0.28 [+0.21, +0.35]	+0.30 [+0.23, +0.37]
Gemma-2-2B	+0.40 [+0.33, +0.47]	+0.11 [+0.04, +0.18]	+0.10 [+0.03, +0.17]	+0.37 [+0.30, +0.43]
OLMo-3-7B	+0.29 [+0.19, +0.38]	−0.30 [−0.40, −0.20]	+0.61 [+0.53, +0.70]	+0.53 [+0.45, +0.62]

Instruct partners (pre-specified). The same readout, window and labelling rule were applied to Llama-3.1-8B-Instruct, Gemma-2-9B-it and Gemma-2-2B-it with their own prefitted lenses and chat-template forced-answer prompts. The criterion allowed the claim that instruction tuning increases the popularity preference only if, for a pair, the instruct difference exceeded the base difference with non-overlapping intervals; this held for no pair.

Table 6: Popularity minus knowledge in the mid-depth window, base and instruct, with 95% intervals. Lenses differ between the two models of a pair, so compare signs, not magnitudes.

	Full readout		Hedge term alone	
Pair	Base	Instruct	Base	Instruct
Llama-3.1-8B	+0.31 [+0.16, +0.46]	−0.35 [−0.71, +0.01]	−0.01 [−0.17, +0.15]	−1.46 [−1.99, −0.97]
Gemma-2-9B	+0.01 [−0.06, +0.08]	−0.98 [−1.17, −0.78]	−0.02 [−0.15, +0.10]	−1.28 [−1.56, −1.00]
Gemma-2-2B	+0.29 [+0.16, +0.42]	+0.15 [+0.07, +0.22]	−0.27 [−0.38, −0.15]	−0.89 [−1.02, −0.76]

Onsets. The onset of an effect curve is the first layer where it reaches half its positive maximum over layers and its 95% interval excludes zero. A model supports the prediction that familiarity acts at a shallower depth than knowledge, our layer-depth analogue of a timing account of feelings of knowing [Koriat and Levy-Sadot, 2001], if the knowledge onset follows the popularity onset with an interval above zero, and contradicts it if the interval lies

below zero. The prediction was not supported (mixed): 1 supporting, 1 contradicting and 6 undecided of the eight base and instruct models.

Table 7: Lens onsets (layer index) of the popularity (F) and knowledge (K) effects, their difference $D = K - F$ with its 95% interval, the outcome, and the mid-depth popularity effect on the full readout. Instruct rows use the instruct models' own lenses. Onsets are integer layers, so bootstrap intervals for D often collapse to a point; they are the inputs to the pre-specified rule, not a precision estimate.

Model	Layers	F onset	K onset	D	Outcome	Mid popularity
Llama-3.1-8B-Instruct	32	17	17	0 [0, 0]	undecided	+1.58
Llama-3.1-8B	32	16	16	0 [0, +1]	undecided	+0.67
Qwen2.5-7B-Instruct	28	20	20	0 [−1, +1]	undecided	+0.21
Gemma-2-9B-it	42	24	24	0 [−2, +1]	undecided	+0.71
Gemma-2-9B	42	24	25	+1 [+1, +2]	supports	+0.39
Gemma-2-2B-it	26	17	15	−2 [−2, 0]	undecided	+0.40
Gemma-2-2B	26	16	16	0 [0, +1]	undecided	+0.40
OLMo-3-7B	32	19	14	−5 [−5, −5]	contradicts	+0.29

E Cross-model agreement (exploratory)

Spearman correlations across questions between Llama's and each other model's forced-answer hedge log-odds and first-token entropies, on questions where both models have the same knowledge label and where they differ (95% bootstrap intervals). The plan, including the rule that hedge agreement and the hedge-minus-entropy gap both have 95% intervals excluding zero, was written before this run, but the point estimates had been seen, so the analysis is post hoc.

	Llama–Qwen	Llama–Gemma	Llama–Mistral
Questions, knowledge differs	313	60	108
Hedge log-odds agreement	+0.11 [−0.00, +0.22]	+0.38 [+0.13, +0.60]	+0.25 [+0.06, +0.42]
Entropy agreement	−0.17 [−0.28, −0.07]	+0.03 [−0.24, +0.28]	−0.07 [−0.27, +0.12]
Gap (hedge − entropy)	+0.28 [+0.14, +0.42]	+0.35 [−0.03, +0.74]	+0.32 [+0.06, +0.57]
Rule	not met	not met	met
Questions, same knowledge	1,784	1,977	2,000
Hedge log-odds agreement	+0.50	+0.38	+0.64
Entropy agreement	+0.46	+0.54	+0.69

F A model-side familiarity measure (negative result)

Pre-specified. We computed the mean per-token surprisal of each subject name under the base Llama-3.1-8B in the context "Here is a name:". It is uncorrelated with log page views (Spearman −0.001). With knowledge and name length held fixed, its coefficient per standard deviation is +0.013 [−0.026, +0.059] on two-judge abstention and +0.07 [+0.05, +0.10] on the regex measure; adding the popularity arm moves the regex coefficient to +0.01 [−0.01, +0.03], although surprisal hardly correlates with the arm (point-biserial +0.02; −0.06 given relation and knowledge), which we have not resolved. Surprisal therefore does not stand in for popularity. Short names made of common words ("New York" as a book title) are a likely reason.

G Pre-specified analyses

Each pre-specified analysis followed a plan written before it ran, stating the measurement, the test and the wording each outcome would license. The plans are in the supplementary material, so each can be checked against the SHA-256 prefix we recorded before its run. The prefixes are self-recorded, not externally timestamped.

Analysis	Plan	SHA-256
Four-family design (Qwen)	`qwen_factorial_criterion.md`	eba35dec
Four-family design (Gemma, Mistral)	`gemma_mistral_factorial_criterion.md`	b53d9a27
One corrected judge	`hedge_judge_criterion.md`	ae0bb807
Two-judge measurement	`hedge_measure_v2_criterion.md`	81acfc9f
Lens study	`jlens_cascade_criterion.md`	c8e04372
Independent labels	`e1_independent_labels.md`	af50f53f
Name surprisal	`e2_name_surprisal.md`	2b69e06c
Lens split	`e3_lens_split.md`	b2b916ce
Steering, held-out dose	`e4_steering_v3.md`; amendment[a]	18c74883; 7bed3826
Third judge	`e5_third_judge.md`	f35aa4d9
Entity recognition	`e6_entity_knowledge.md`	937b46e3
Regraded labels, strict declines	`e7_label_and_decline_audit.md`	1498be91
Within-level steering	`e8_steering_within_level.md`	05474954
Permissive prompt	`e9_permissive_prompt.md`	f6459163
Instruct lens	`e10_instruct_lens.md`	6af179ce

[a]Adds the third judge as a secondary scorer; its internal date is one day later than its recorded hash time.

The guessability, namesake, name-spelling, free-reply, entropy-control and label-agreement checks, the alias audit and the cross-model agreement were decided after seeing data and are post hoc.